

6	(a) <i>either</i> constant speed parallel to plate or accelerated motion/force normal to plate/in direction field so not circular	B1 A0	[1]
	(b) (i) direction of force due to magnetic field opposite to that due to electric field magnetic field into plane of page	B1 B1	[2]
	(ii) force due to magnetic field = force due to electric field	B1	
	$Bqv = qE$		
	$B = E / v$	C1	
	$= (2.8 \times 10^4) / (4.7 \times 10^5)$		
	$= 6.0 \times 10^{-2} \text{ T}$	A1	[3]
	(c) (i) no change/not deviated	B1	[1]
	(ii) deviated upwards	B1	[1]
	(iii) no change/not deviated	B1	[1]